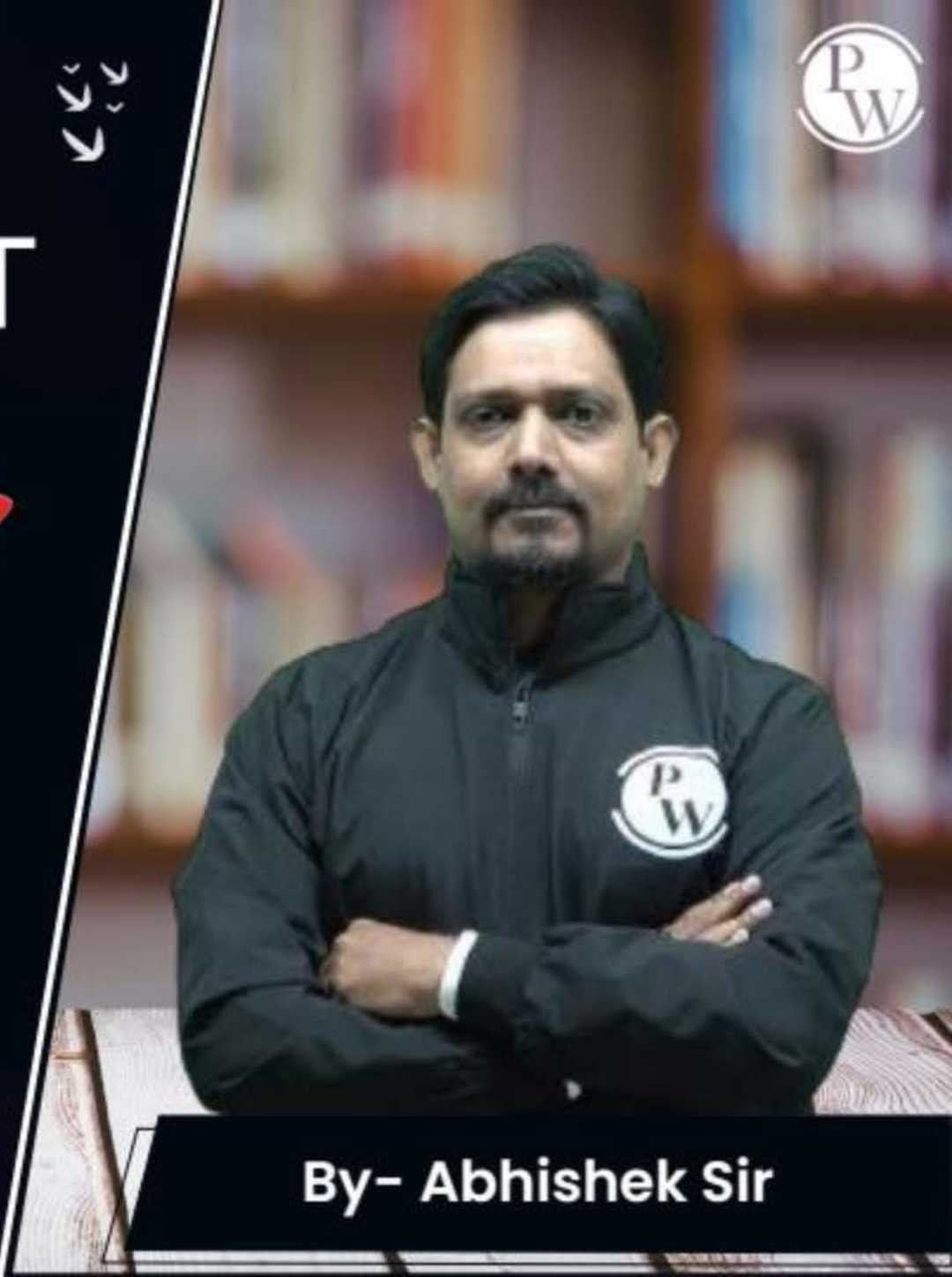
Computer Science & IT



C Programming

Control Flow Statement

Lecture No. 02



By- Abhishek Sir

Recap of Previous Lecture



- Topic — Ternary operator
- Topic — if-else
- Topic — if else if
- Topic — Nested if
- Topic

Slide

Topics to be Covered



- Topic — Switch statement
- Topic — for Loop
- Topic
- Topic
- Topic

Slide



Switch statements

Switch statement allows a value to be test for equality against list of values.

* each list value is called case
* case constant expression
* if None of the case matches the default case executes
* break & default is optinal.

Slide



Switch statements

```
Switch (expression) {
    case constant expression : statements; break;
    case constant expression : statements; break;
    .
    .
    .
    defaut : statements; break;
}
```

Slide



Application

Menu driven program



Switch Syntax

```
switch(expression){
    case constant-expression  :
       statement(s);
       break; //optional
    case constant-expression  :
       statement(s);
       break; //optional

    // you can have any number of case statements.
    default : //Optional
       statement(s);
}
```



If No break statement

* if break is Not present then all Subsequent (fall through) are evaluated to true.

* break transfer the control outside of Switch block.

Slide



Question

```
#include <stdio.h>
      int main(){
       int a = 5;
        switch(a){
                  case 4: printf("%d",5);
                  case 5: printf("%d",5);
                  case 6: printf("%d",6);
        }
      }
```

output

5 6

reason :

No break present in case 5

Slide



Question

```
#include <stdio.h>
    int main(){
     int a = 5;
      switch(a-1){
            case a: printf("%d",5);
            case 5: printf("%d",5);
            case 6: printf("%d",6);
      }
    }
```

expression

a is variable

output :

Error

reason

case a

↑

a is Not constant

a is a variable

Slide



Question

```
#include <stdio.h>
int main(){

    switch(13/4){
        case 4: printf("%d",4); break ;
        case 3: printf("%d",2); break ;
        case 5: printf("%d",5); break ;
}
    }
```

$13/4 = \frac{int}{int} = 3$

int

output

2

reason :

Case 3 match

int/int = int

Slide



switch(expression)

```
#include <stdio.h>
      int main(){
       int a = 5;
        switch(a-1){
                  case 4: printf("%d",5);
                  case 5: printf("%d",5);
                  case 6: printf("%d",6);
       }
      }
```

Output

Reason



switch(expression)

Output

```
#include <stdio.h>
int main(){
        switch(13/4){
        case 3: printf("%d",4); break ;
        case 4: printf("%d",2); break ;
        case 5: printf("%d",5); break ;
        }
    }
```

Reason



switch(expression)

```c
#include <stdio.h>
int main(){
        switch(13/4.0){
        case 3: printf("%d",4); break ;
        case 4: printf("%d",2); break ;
        case 5: printf("%d",5); break ;
        }
    }
```

int / float : 13 / 4.0 = 3.25

Output

Error

Reason

13/4.0 = 3.25

float is Not allowed in Switch expression

Slide



switch(expression)

```
#include <stdio.h>
int main(){
      switch('d'){
       case 'a': printf("%d",4); break ;
       case 'b': printf("%d",2); break ;
       default: printf("None");
       case 'c': printf("%d",5); break ;
      }
}
```

Output

None 5

Reason

1 No case matches hence default will execute

2. No break present

Hence case 'c' will also execute.



switch(expression)

```
#include <stdio.h>
int main(){
        switch(4){
        case 2: printf("%d",4); break ;
        case 1+1: printf("%d",2); break ;
        default: printf("None");
    }
}
```

OK.

Output

Error

Reason

Case 2 :

Case 1+1 :

duplicate case is Not allowed.



switch(expression)

```
#include <stdio.h>
int main(){
        int a=2;
         switch(4){
         case 2: printf("%d",4); break ;
         case a: printf("%d",2); break ;
         default: printf("None");
      }
}
```

Output

Reason

Slide



GATE 2012

What will be the output of the following C program segment?

```
Char inChar = 'A' ;
switch (inChar ) {
case 'A' : printf ("Choice A\ n");
      case 'B' :
      case 'C' : printf("Choice B");
      case 'D' :
      case 'E' :
      default  : printf ("No Choice") ; }
```

(A) No choice
(B) Choice A
(C) Choice A
Choice B No choice
(D) Program gives no output as it is erroneous



GATE 2012

\t tab

What will be the output of the following C program segment?

```
Char inChar = 'A' ;
switch (inChar ) {
case 'A' : printf ("Choice A\ n");
      case 'B' :
      case 'C' : printf("Choice B");
      case 'D' :
      case 'E' :
      default  : printf ("No Choice") ; }
```

→ Newline

(A) No choice
(B) Choice A
(C) Choice A
Choice B No choice
(D) Program gives no output as it is erroneous

Choice A
Choic B No choice

Slide



Question

What will be the output of the following C program segment?

```
#include <stdio.h>
      int main(){
       int a = 80;
        switch(a){
                  case 80: a = a+10;
                  case 5: a++;
                  default : a = a>>2;
        }
      printf("%d",a);

}
```

(A) 20

(B) 21

(C) 22

(D) 23

80

a = 90

a++; = 91

a>>2 : $\left\lfloor \frac{91}{4} \right\rfloor$ = 22

Slide



Question

What will be the output of the following C program segment?

```
#include <stdio.h>
      int main(){
       int a = 80;
        switch(-12%45+36/9/2*16+60){
                  case 80: a = a+10;
                  case 5: a++;
                  default : a = a>>2;
        }
      printf("%d",a);
}
```

80

(A) 20

(B) 21

(C) 22

(D) 23

$-12\%45+36/9/2*16+60$

$-12+4/2*16+60$

$-12+2*16+60 = -12+32+60$

$= 20+60 = 80$

Slide



Question

What will be the output of the following C program segment?

```
#include <stdio.h>
     int main(){
      int a = 5, b=24, c = 10;
       switch(28/5){
                case 5: a = a&b;
                case 6: b = b^c; break;
                default : a = a>>2;
       }
     printf("%d",a+b+c);
}
```

a = 000101

b = 011000

10 = 0000000

a = 0

b = 18

0 + 18 + 10 = 28

c = 001010

```
  011000
  001010
--------
  010010
```

(A) 20

(B) 21

(C) 22

(D) 28

Slide



Question

```
int main()
{
    int a =50;
    switch(a)
    {
        default: a=45;
        case 49: a++;
        case 50: a--;
        case 51: a =a+1;
    }
    printf("%d",a);
}
```

(A) 51

(B) 45

(C) 49

(D) Error

Slide



Question

```
int main()
{
    int a =50;
    switch(a)
    {
        default: a=45;
        case 49: a++;
        case 50: a--;
        case 51: a =a+1;
    }
    printf("%d",a);
}
```

(A) 51

(B) 45

(C) 49

(D) Error

Slide



Question

```
int main(){
      char check = 'a';
      switch (check){
      case 1'a' || 1: printf("Gates ");
      case 1'b' || 2: printf("Quiz ");
            break;
      default: printf("GatesQuiz");

      }
      return 0;
      }
```

logical expression

(a) Gates

(b) Gates Quiz

(c) Gates Quiz GatesQuiz

(d) Compile-time error

duplicate case

Not allowed

Slide



Question

```
#include<stdio.h>
int main(){
      int check = 4;
      char ch ='0';
      switch (-16%2+4){
            case 2*13/4+-8%3: ch+=3 ;
            case 'b' || 2: ch+2;
                  break;
            default: ch-=4;

      }
      printf("%d", ch);
      return 0;
}
```

(a) '0'

(b) 51

(c) 53

(d) Compile-time error



Position of default does not matter

Slide



What is Iterative Statement ?

Iterative statement

If we want to execute a block of code multiple times then iterative statement used

1. for Loop
2. while
3. do while

Slide



What is Iterative Statement ?

Print My name 100 times

repeat printf 100 times

Slide



Second Way is loop

Print My name 100 times

Slide



For loop: Syntax

Print My name 100 times

```
for (expr1; expr2; expr3) {

}
```

expr1 is assignment / function call

expr2 is Conditional expression expression or function

expr3 is assignment Increament/Decrement or function call

Slide



For loop: Syntax

Different range of numbers

```
int i;
for(i =1;i<=10;i++)
    printf("Name");
```

1 → one time

2 → Every time condition checked

3 → Iteration every time

4

order of execution

1. assign
2. condition check
4. Block of code will execute
3. Increment

Slide



For loop: Syntax

Different range of numbers

```
int i;
for(i =1;i<=10;i++)

    printf("Name");
```

i = 1
i = 2
i = 3
:
:
i = 10

10 timmes
Name

Slide



For loop: Syntax

Different range of numbers

LB UB

```
for(i =11; i<=21; i++)
    printf("Name");
```

UB - LB + 1 ⇐

21 - 11 + 1 = 11

i=5 ; i<=8 ; i++

i=5
i=6
i=7
i=8

} 4 times.

Slide



For loop: Syntax

Different range of numbers

```
for(i =-10; i<=21; i++)
        LB      UB
    printf("Name");
```

UB - LB + 1

21 - (-10) + 1 = 31 + 1 = 32

Slide



For loop: Syntax

Different range of numbers

```
for(i =-10; i<=21; i++)
    printf("Name");
```



For loop: Syntax

Different range of numbers

−10 — 20

```
for(i =-10; i<21; i++)

    printf("Name");
```

UB - LB + 1

20 - (-10) + 1 = 31

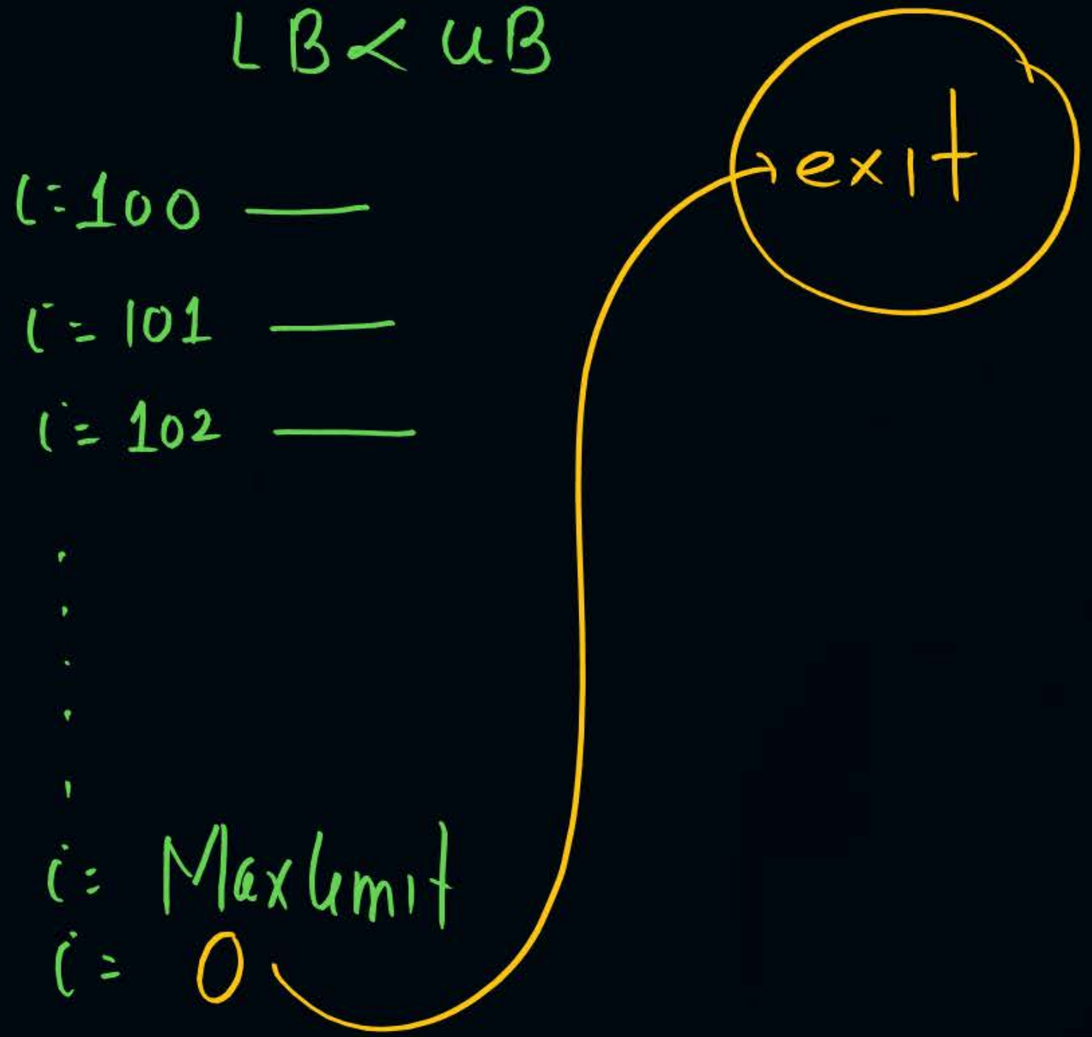
Slide



For loop: Syntax

Different range of numbers

```
for(i =100; i>=1; i++)
    printf("Name");
```

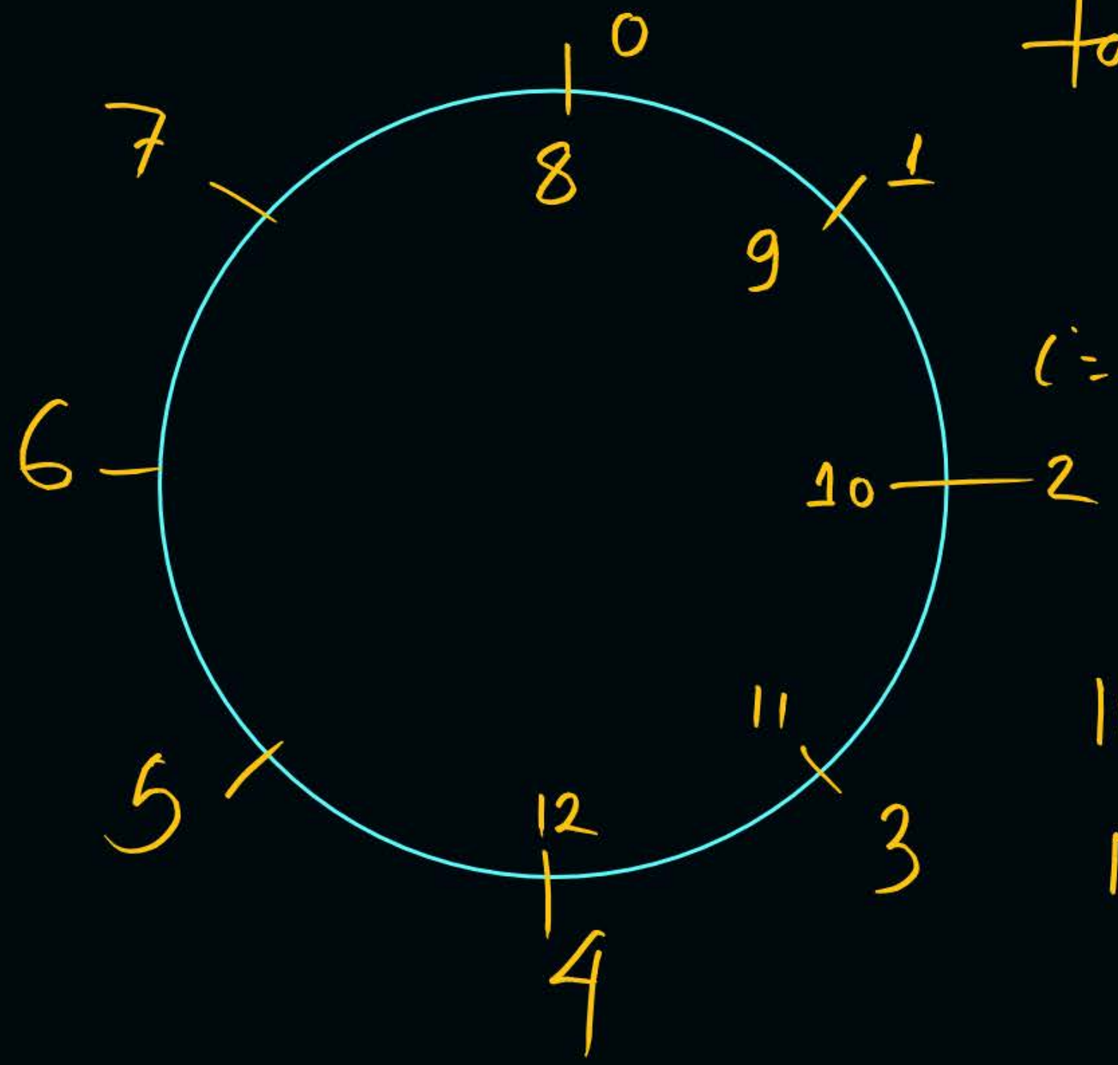


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e.g.

3 bit

000
001
010
011
100
101
110
111
(1)000
1001



for explanation purpose 3bits

i = 12

will print

12%2^3 ← Size of bits

12%8 = 4

```
short int a=32770;
printf("%d", a);
```

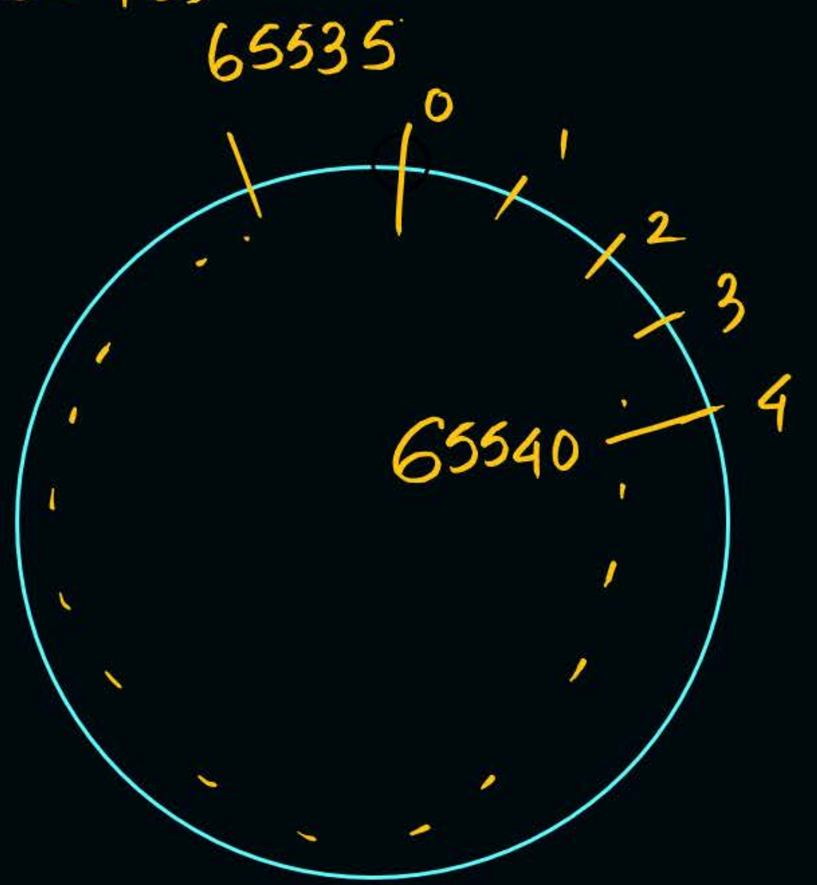
-2^{15} ——— 2^{15}

-32768 —— 1 —— 32767

$0 \text{ —— } 2^{16}-1$

short unsinged int a = 65540;

printf("%u", a);



unsigned overflow

short size
= 2B
= 16

65540 % 65536 = 4

65540 % 2^{size}

Q1

```
short unsigned int a = 70000;
printf("%u", a);
```

70,000

65536

1 1 1 1

04464



2 mins Summary

Topic — Switch statement

Topic — for loop

Topic — unsigned overflow

Topic

Topic

Slide

THANK - YOU